

Algorithmic Pseudocode Sample 52

Source-backed Image2Struct algorithm sample

1 Algorithm

This sample contains algorithmic pseudocode extracted from a source-backed LaTeX benchmark dataset. It is wrapped in a minimal article document for pdfLaTeX validation.

Algorithm 1 Source-backed algorithmic procedure

Require: $n = 2, N \in \mathbb{N}$ sufficiently large

Require: X_{1i} are independent for $i = 0, 1, \dots, k - 1$

Require: $X_{2i} = 1 + r, r > -1$ for $i = 0, 1, \dots, k - 1$

Require: $\pi_i = (q_i(\widehat{W}_i, A_i), 1 - q_i(\widehat{W}_i, A_i))$ for $i = 0, 1, \dots, k - 1$

$l \leftarrow 0$

▷ initialize l

while $l \leq N$ **do**

$l \leftarrow l + 1$

$i \leftarrow 0$

▷ initialize i

$I \leftarrow 1$

▷ initialize I_i

$A \leftarrow A_0$

▷ initialize A_i

$\widehat{W} \leftarrow P$

▷ initialize $\widehat{W}_i$

while $i \leq k$ **do**

$i \leftarrow i + 1$

X is a realization of $X_{1,i-1}$

$Y \leftarrow q_{i-1}(\overline{W}, A) \cdot X + (1 - q_{i-1}(\overline{W}, A)) \cdot (1 + r)$

B^j is a realization of B_{i-1}^j for $j = 1, 2, \dots, A$

$I \leftarrow I - \sum_{j=1}^I B^j$

$A \leftarrow A - \sum_{j=1}^A B^j$

$\widehat{W} \leftarrow Y\widehat{W} - IAw_i$

▷ computes $\widehat{W}_i$

end while

$b_l \leftarrow \begin{cases} 1, & \widehat{W} \geq 0 \\ 0, & \text{otherwise} \end{cases}$

end while

$\mathbb{P}(\widehat{W}_k \geq w) \leftarrow \frac{1}{N} \sum_{l=1}^N b_l$

return $\mathbb{P}(\widehat{W}_k \geq w)$
